

Marcos Arnold

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EDUCATION

Colorado College

Colorado Springs, CO

Bachelor of Arts, Double Major in Computer Science and Sociology

May 2027

- Relevant Coursework: Data Structures and Algorithms, Software Design, Computer Organization, Number Theory

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C/C++, HTML, CSS, GIS

Frameworks: React.js, Node.js, Django, MongoDB, REST API, HTMX, Tailwind CSS, PyTorch, FastAPI

Developer Tools: Git, Github Actions, Apache Maven, OpenAI, Claude, Figma, Postman, AWS, Firebase, Docker

EXPERIENCE

Human-Computer Interaction (HCI) Research Assistant

November 2025 – Present

Colorado College Department of Computer Science

Colorado Springs, CO

- Conducting literature review on accessibility and cognitive disabilities, synthesizing 15+ ACM HCI papers to inform system design decisions and feature requirements for tools supporting neurodivergent users.
- Integrating Hugging Face diffusion models with Meshy APIs to generate 100+ images across 20+ prompts, supporting 2D-to-3D conversion for accessibility-focused assistive object design.

Software Development Intern

May 2025 – October 2025

Colorado College Office of Information Technology

Colorado Springs, CO

- Architected a full-stack Django and HTMX application that centralized tracking for 23 institutional applications across 19 business processes, modeling a hierarchical service flow from division to department, service area, business function, and application to improve cross-department coordination.
- Designed and implemented a PostgreSQL relational database managing 1K+ records, improving data integrity by eliminating duplicate entries and reducing cross campus reporting query time by 40%.

TECH PROJECTS

Hue Hero | *Java, Java Swing, SQL, Apache Maven CI/CD, Figma, Image Processing*

December 2025

- Engineered a desktop color-by-number pixel art game used by 20+ players, converting images into interactive grids and supporting 50+ leaderboard entries through a MySQL backend, with Maven CI/CD using GitHub Actions.

Twitter Community Detection | *Python, NetworkX, Pandas, Leiden, Infomap*

December 2025

- Analyzed a Twitter network of 81,306 users from the Stanford SNAP dataset with Leiden and Infomap algorithms to measure information flow and quantify echo chamber effects, producing visualizations of community structures.

CashCow | *React, TypeScript, Fetch.AI, FastAPI, BrightData MCP, Anthropic Claude*

October 2025

- Built an AI-powered Chrome extension detecting checkout pages across 100+ merchants, classifying in real-time with Fetch.AI and Claude, and recommending optimal credit card rewards to maximize cashback and points.

Snake AI | *Python, Pygame, NumPy, Matplotlib, Jupyter Notebook*

May 2025

- Developed an AI-powered Snake game in Python and Pygame, implementing Greedy, Dijkstra, and A* pathfinding algorithm to optimize real-time navigation in dynamic environments, achieving improved decision-making efficiency and obstacle avoidance.

ADDITIONAL EXPERIENCES

Leadership Team

February 2025 – Present

Colorado College Coding Club

Colorado Springs, CO

- Mentoring 60+ students through workshops and community events, helping members achieve personal, academic, and professional coding goals.
- Founded Tiger HaCCs, the inaugural interdisciplinary campus hackathon, coordinating 40+ participants, securing sponsorships, and leading hands-on software development projects with Career Center and Math/CS Department.

Webmaster Assistant

September 2025 – Present

The Sounds of Colorado College (SOCC) Radio

Colorado Springs, CO

- Developing and maintaining the SOCC Radio website using WordPress, Cargo, HTML, and CSS, supporting 500+ weekly visitors by implementing content updates, multimedia integrations, and accessibility improvements.